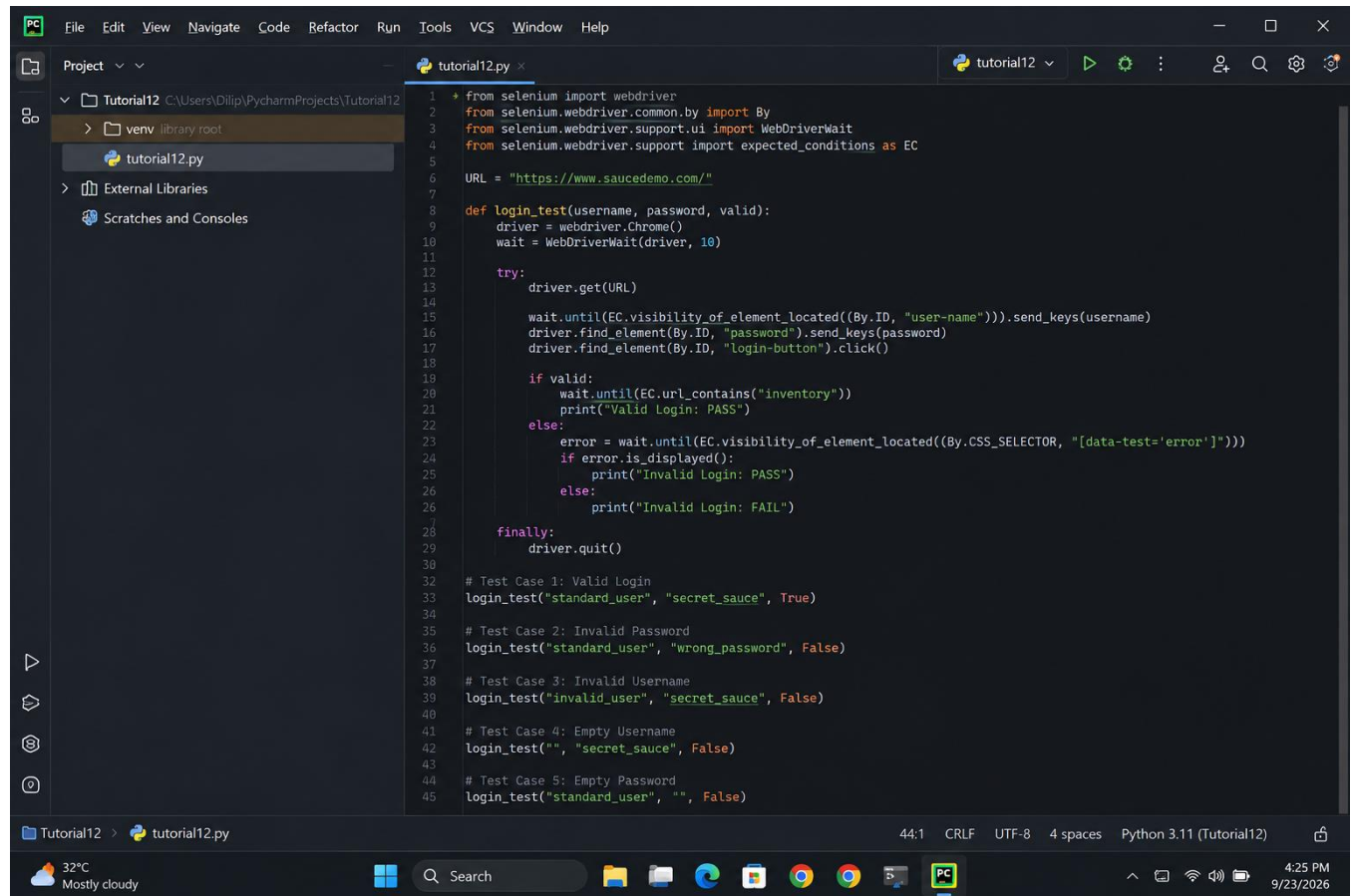


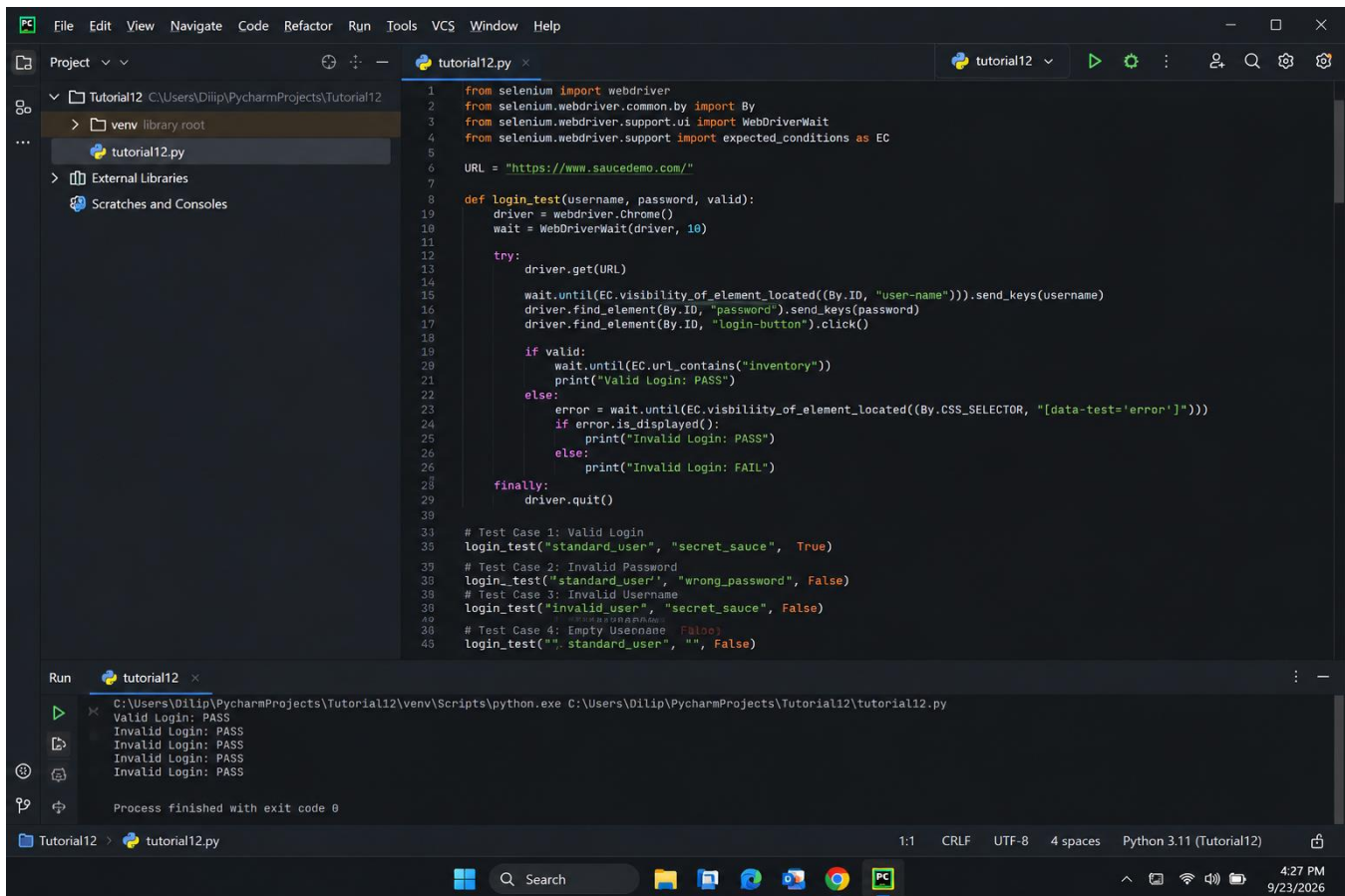
PyCharm Setup / Source Code

Created a Python file named tutorial12.py in PyCharm and use the Selenium WebDriver code provided for the practical. The source-code screenshot can be pasted below.

OUTPUT / SCREENSHOT



```
1 from selenium import webdriver
2 from selenium.webdriver.common.by import By
3 from selenium.webdriver.support.ui import WebDriverWait
4 from selenium.webdriver.support import expected_conditions as EC
5
6 URL = "https://www.saucedemo.com/"
7
8 def login_test(username, password, valid):
9     driver = webdriver.Chrome()
10    wait = WebDriverWait(driver, 10)
11
12    try:
13        driver.get(URL)
14
15        wait.until(EC.visibility_of_element_located((By.ID, "user-name"))).send_keys(username)
16        driver.find_element(By.ID, "password").send_keys(password)
17        driver.find_element(By.ID, "login-button").click()
18
19        if valid:
20            wait.until(EC.url_contains("inventory"))
21            print("Valid Login: PASS")
22        else:
23            error = wait.until(EC.visibility_of_element_located((By.CSS_SELECTOR, "[data-test='error']")))
24            if error.is_displayed():
25                print("Invalid Login: PASS")
26            else:
27                print("Invalid Login: FAIL")
28
29    finally:
30        driver.quit()
31
32 # Test Case 1: Valid Login
33 login_test("standard_user", "secret_sauce", True)
34
35 # Test Case 2: Invalid Password
36 login_test("standard_user", "wrong_password", False)
37
38 # Test Case 3: Invalid Username
39 login_test("invalid_user", "secret_sauce", False)
40
41 # Test Case 4: Empty Username
42 login_test("", "secret_sauce", False)
43
44 # Test Case 5: Empty Password
45 login_test("standard_user", "", False)
```



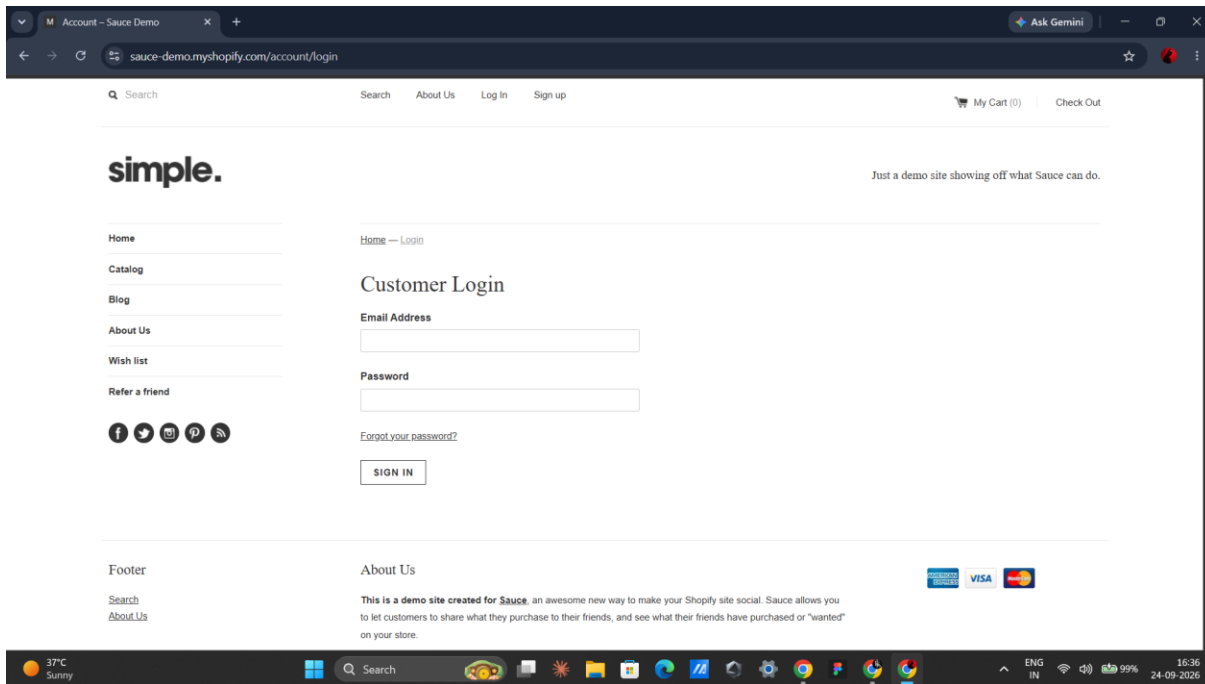
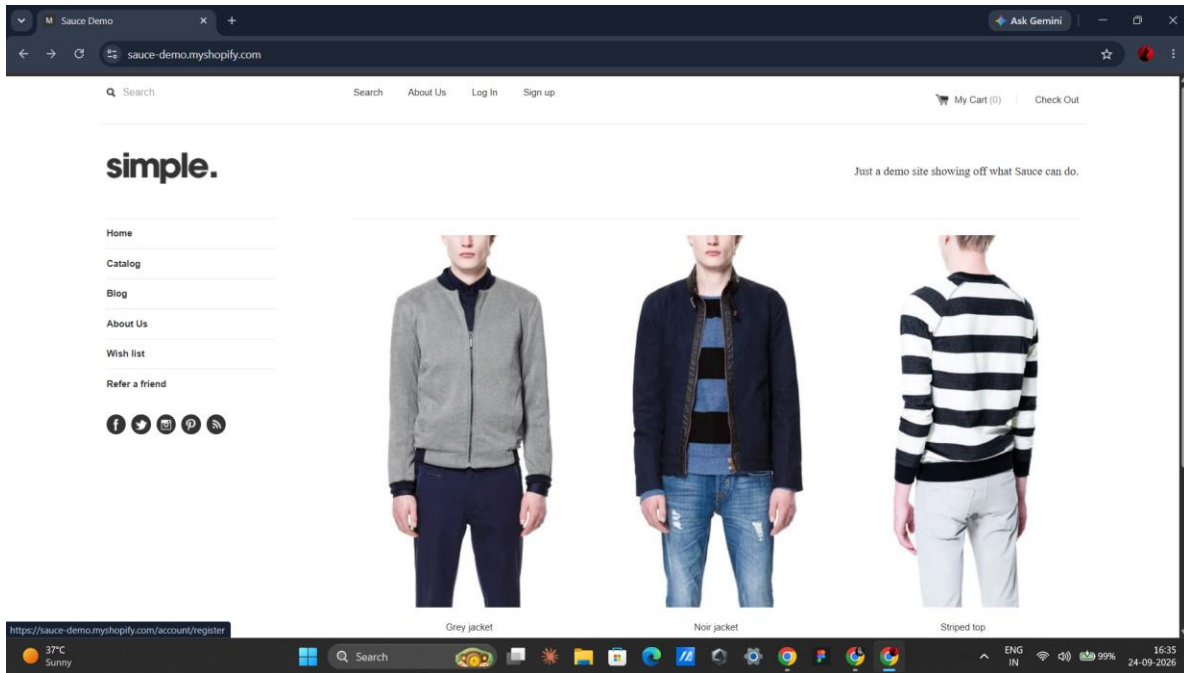
Scenario 1: Valid Login

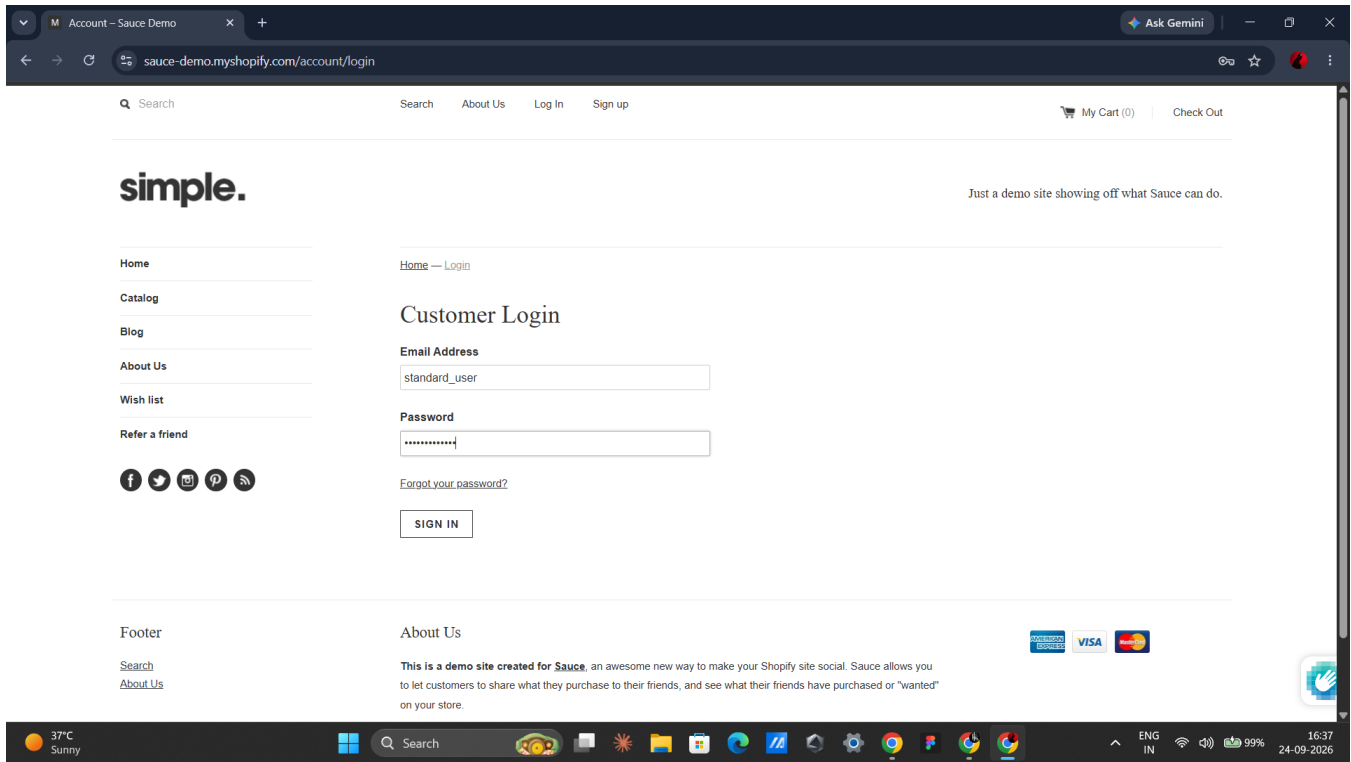
Test Steps

1. Open the SauceDemo Web Application.
2. Enter username: standard_user.
3. Enter password: secret_sauce.
4. Click the Login button.
5. Verify that the Products page is displayed.

Expected Result: User should be redirected to the Products page.

OUTPUT / SCREENSHOT





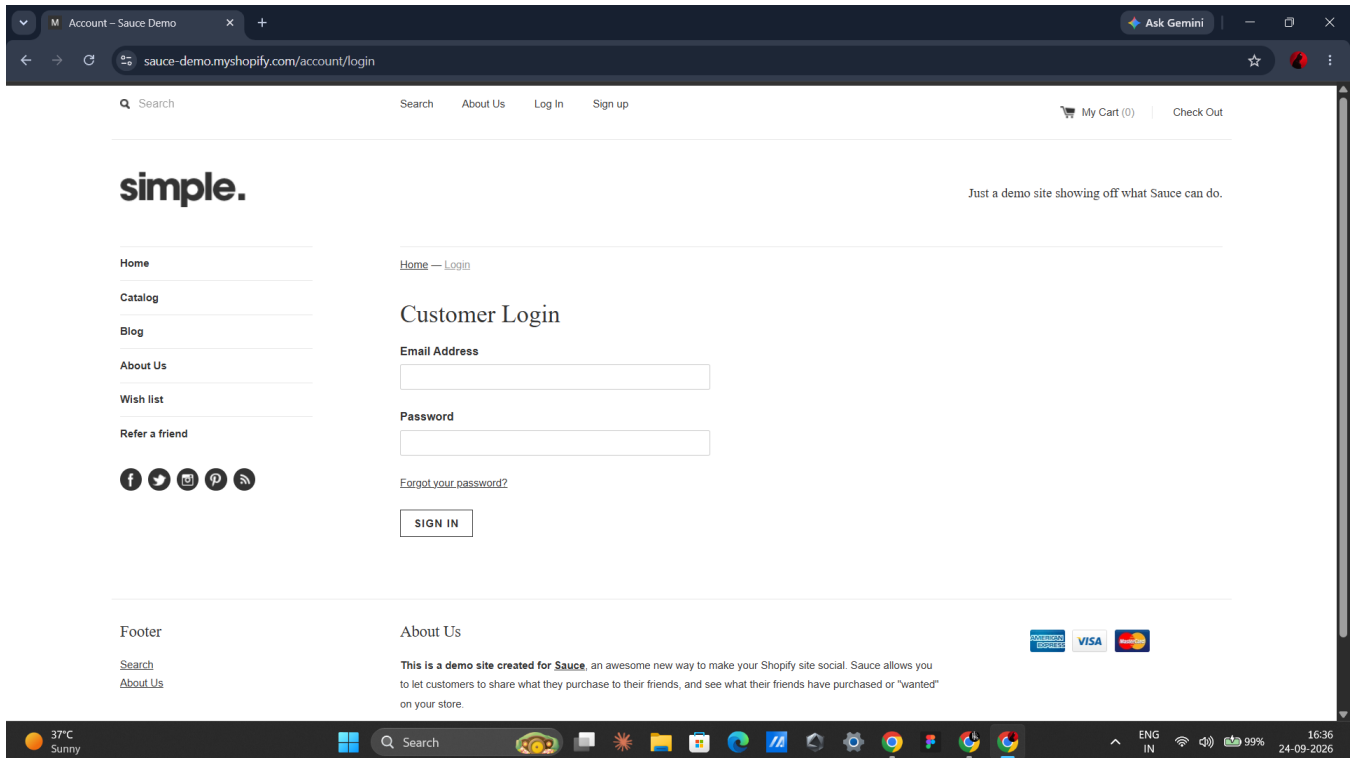
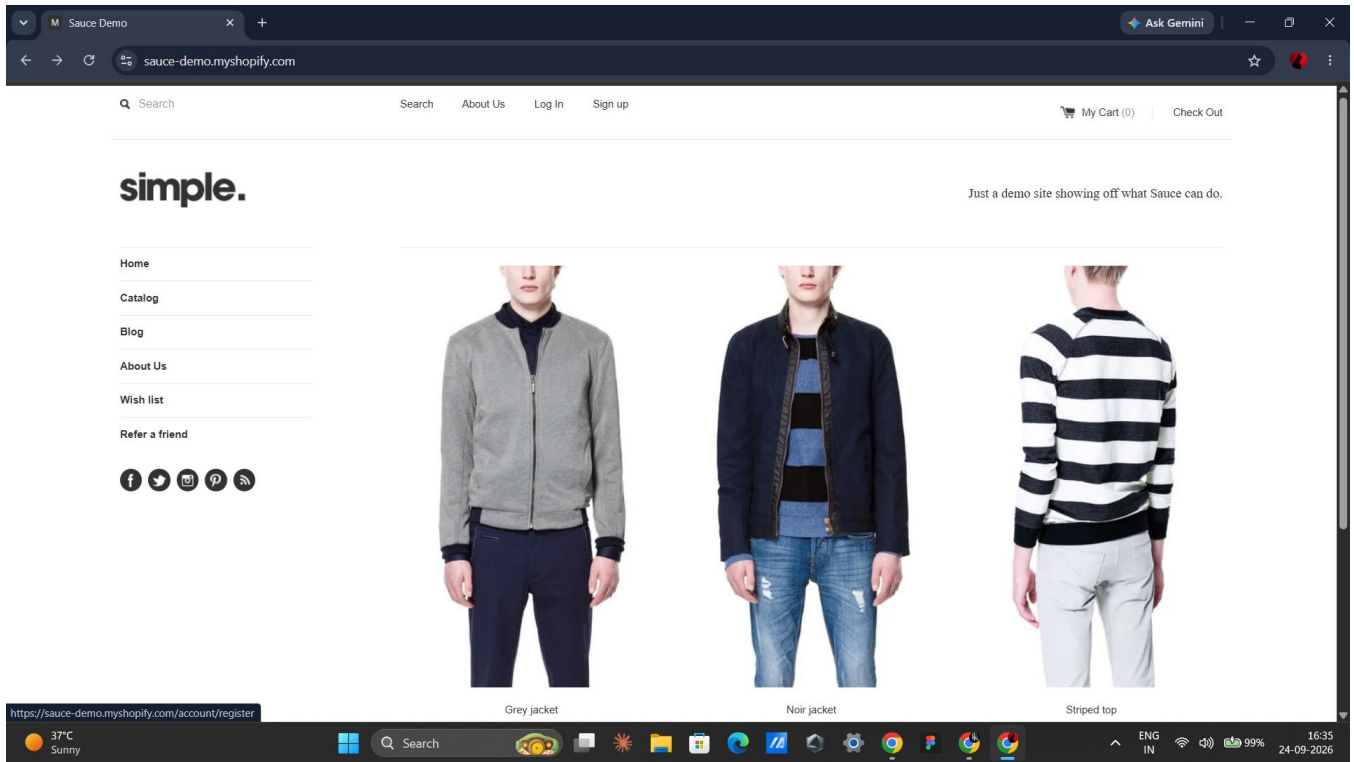
Scenario 2: Invalid Password

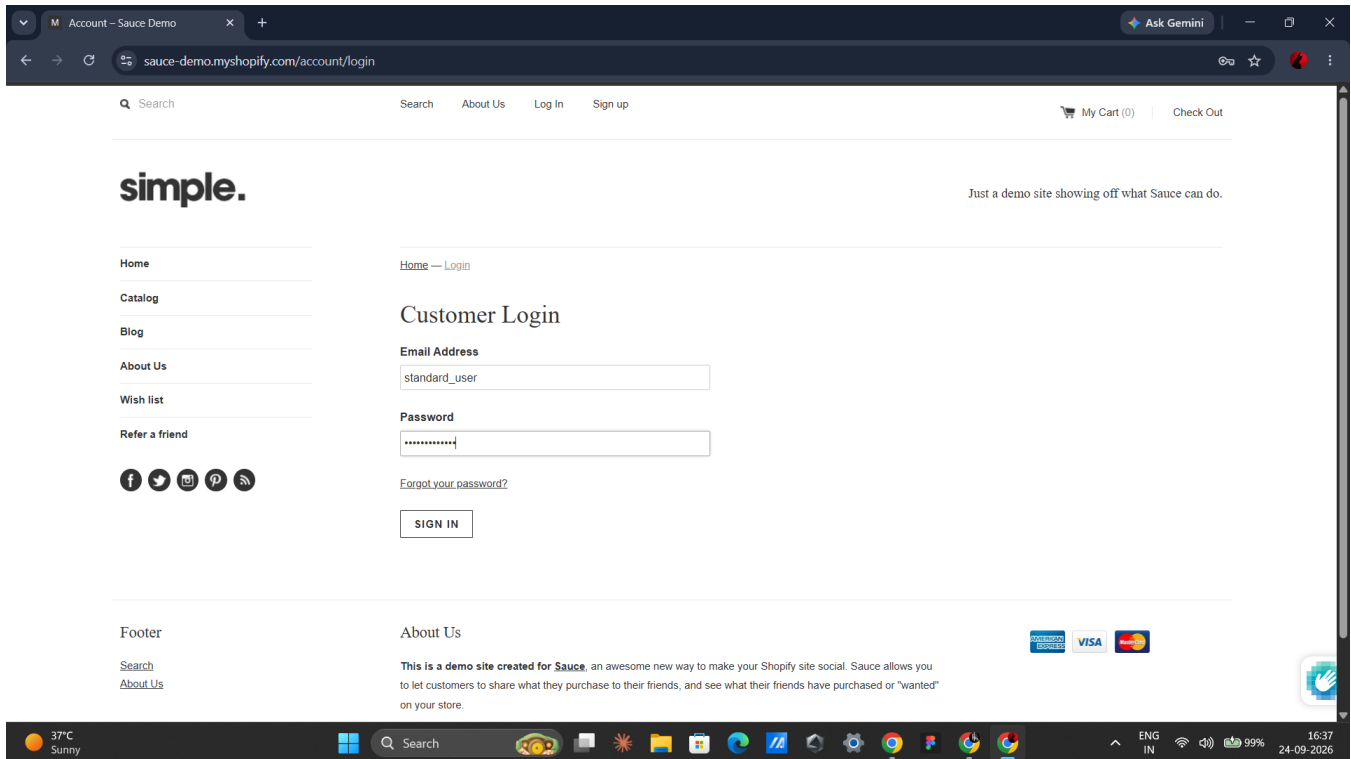
Test Steps

1. Open the SauceDemo Web Application.
2. Enter username: standard_user.
3. Enter an incorrect password.
4. Click the Login button.
5. Verify that an error message is displayed.

Expected Result: An appropriate login error or validation message should be displayed.

OUTPUT / SCREENSHOT





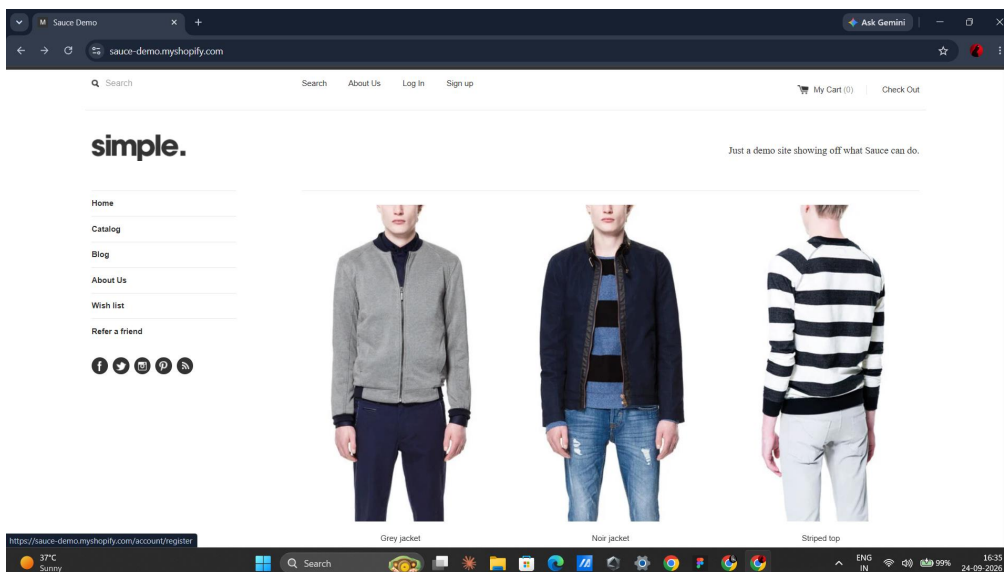
Scenario 3: Invalid Username

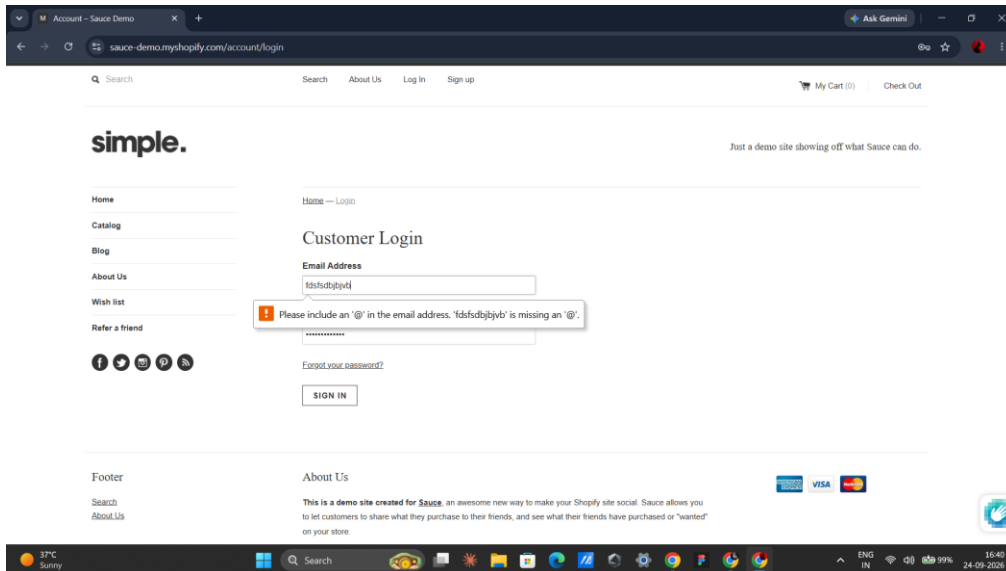
Test Steps

1. Open the SauceDemo Web Application.
2. Enter an invalid username.
3. Enter password: secret_sauce.
4. Click the Login button.
5. Verify that an error message is displayed.

Expected Result: An appropriate login error or validation message should be displayed.

OUTPUT / SCREENSHOT





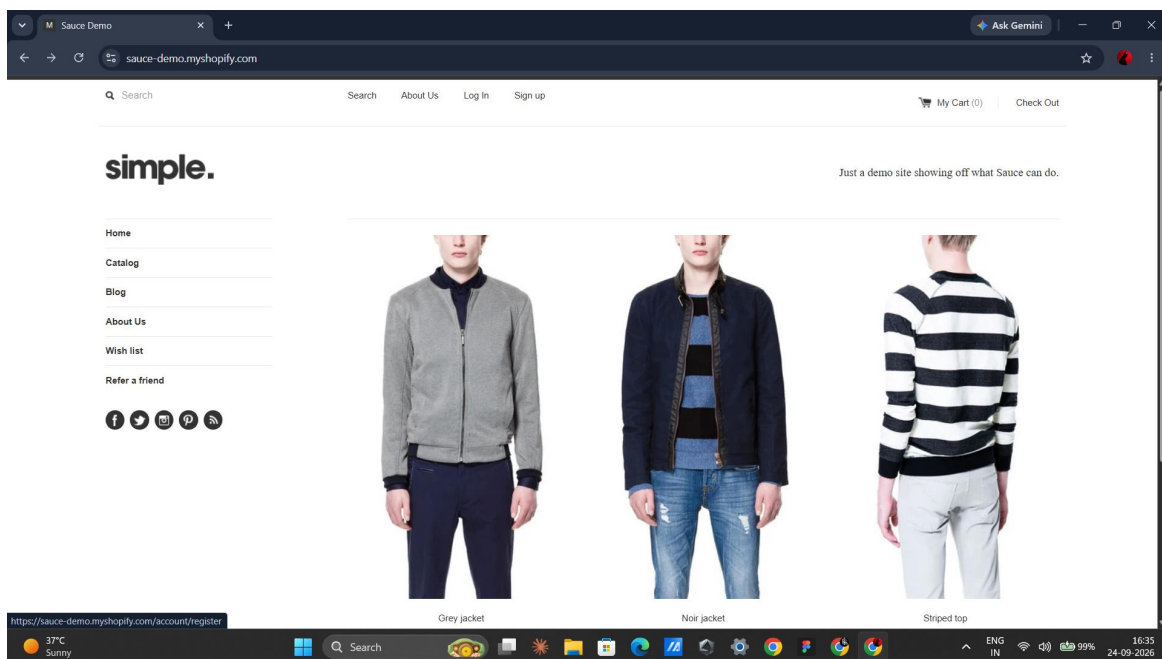
Scenario 4: Empty Username

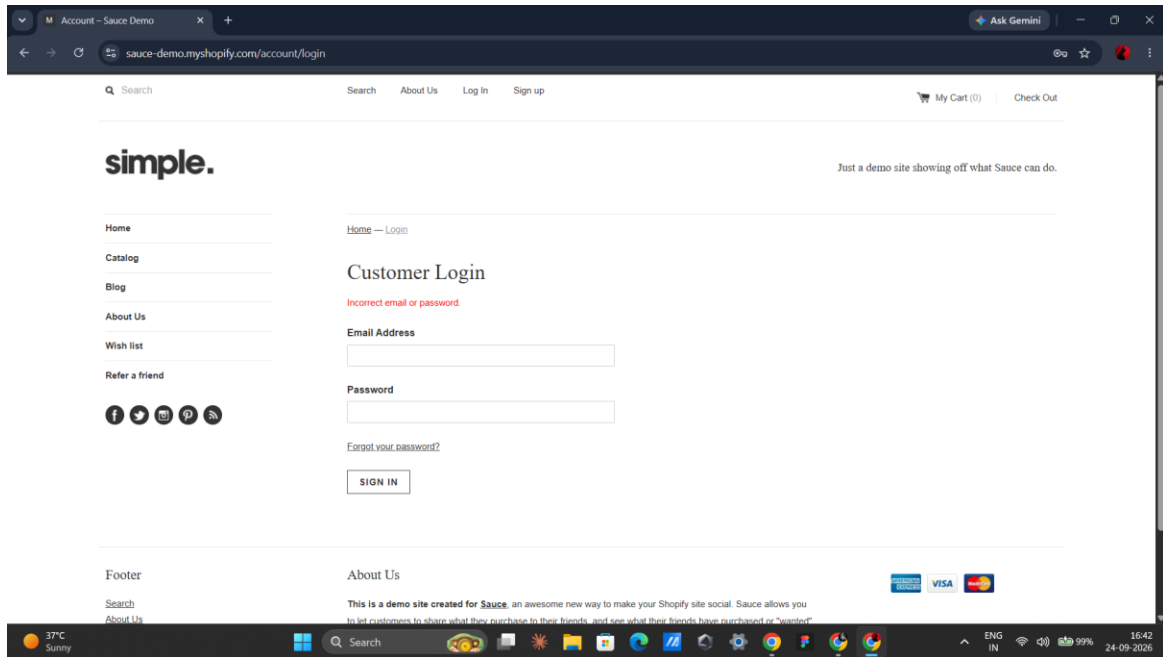
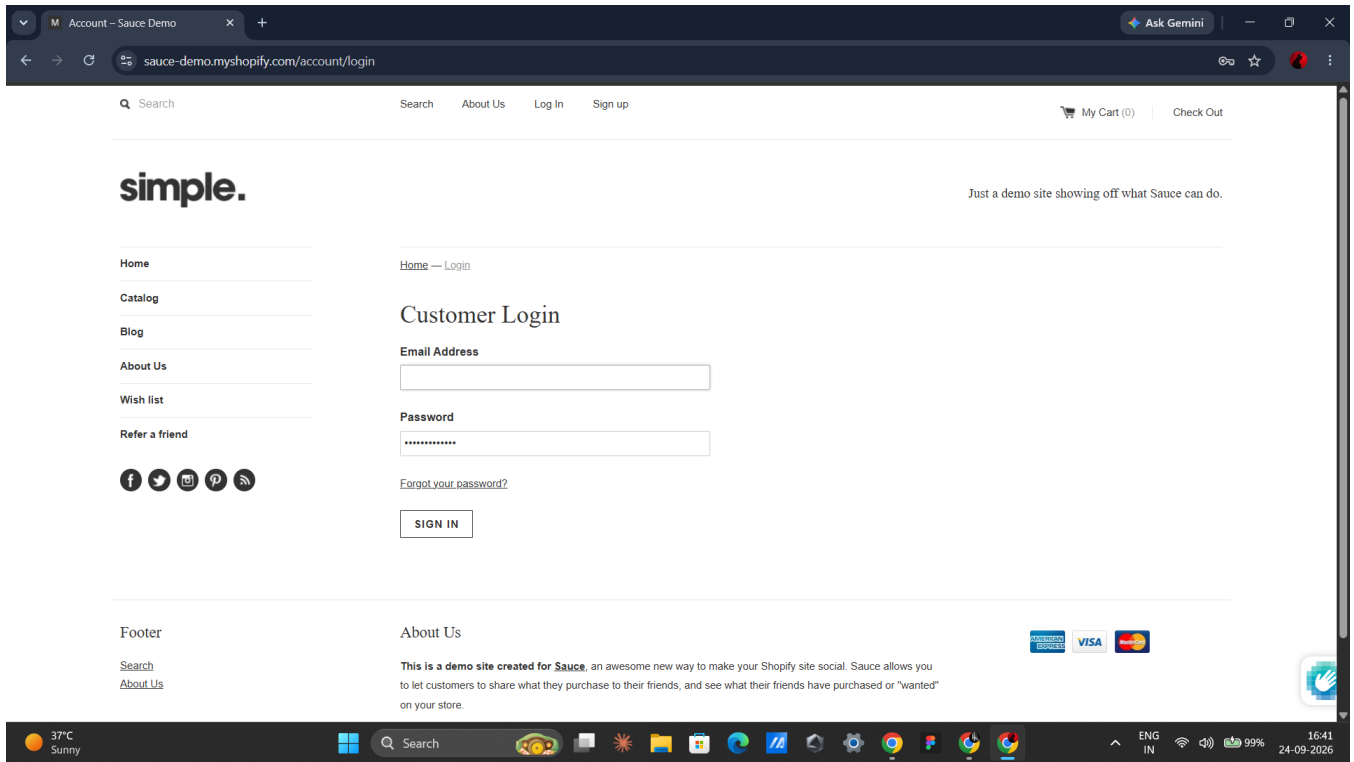
Test Steps

1. Open the SauceDemo Web Application.
2. Leave the username field blank.
3. Enter password: secret_sauce.
4. Click the Login button.
5. Verify that a username-required validation message is displayed.

Expected Result: An appropriate login error or validation message should be displayed.

OUTPUT / SCREENSHOT





Conclusion / Outcome

Python Selenium automation testing was prepared for the SauceDemo Login and Authentication module. The practical covers four scenarios: valid login, invalid password, invalid username, and empty username. The source-code screenshot and scenario screenshots can be added in the blank spaces provided.